KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 25 · CHECKPOINT

Checkpoint 5 — Charting the Course Review — Teacher Lesson Plan & Answer Key

Learning objectives: Students show mastery of Unit 5 (Weeks 21–24): the coordinate plane and number patterns, measurement conversions, line plots with fractions, and classifying two-dimensional figures and symmetry, plus text structures and signal words, firsthand and secondhand accounts, figurative language and word choice, and character change and inference. Students also solve multi-step problems, judge explanations, find mistakes, edit sentences, and write a short firsthand paragraph that combines the unit's writing skills.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · scrap paper · ruler if available · printed Checkpoint Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: the four Unit 5 math settings (Quinlan's trail map in Yellowstone National Park, Solveig's tugboat in the Duluth, Minnesota, harbor, Vikram's tide pools at Haystack Rock in Cannon Beach, Oregon, Ximena's bridge design studio in Pittsburgh, Pennsylvania) and the reading texts about Mesa Verde, the first Moon landing, a Nashville fiddle camp and Sitka, Alaska. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, the answer plus the reason appear after the 2nd miss, and every section has a week tag so a missed item points to the week to review.
2. Slide 1 — Pencil Break (Pre): the student opens or prints the Checkpoint Pre/Post sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W25.pdf) and completes Page 1 (Checkpoint 5 Warm-Up) off screen, with a ruler if available. Do not help with answers yet; it shows where the student is starting.
3. Slide 2 — Week 21 Math: The Coordinate Plane, Part 1 (5 pts: 3 + 2). Each skill slide has a one-line reminder at the top; if a student misses an item, point back to it.
4. Slide 3 — Week 21 Math: Patterns, Graphs, and Map Scale, Part 2 (2 pts): a two-rule pattern or a map-scale distance item.
5. Slide 4 — Week 21 Reading: Text Structures, Part 1 (5 pts: 3 + 2), with a short co-op report or garden journal entry in each set. Ask what the WHOLE text is mostly doing.
6. Slide 5 — Week 21 Reading: Signal-Word Commas, Part 2 (2 pts): fix capitals and the comma after a signal word, a because part, or before so; checked exactly.
7. Slide 6 — Week 22 Math: Measurement Conversions, Part 1 (5 pts: 3 + 2). Tip: bigger unit to smaller unit means multiply.
8. Slide 7 — Week 22 Math: Smaller Units to Bigger Units, Part 2 (2 pts): divide, then write the remainder as a mixed measurement or a fraction of the bigger unit.
9. Slide 8 — Week 22 Reading: Firsthand and Secondhand Accounts, Part 1 (5 pts: 3 + 2), with two short accounts of one event (or one interview) in each set. Ask: firsthand about WHAT?
10. Slide 9 — Week 22 Reading: Dates, Quotations, and Pronouns, Part 2 (2 pts): commas in dates, a speaker-tag comma, or I versus me; checked exactly.

11. Slide 10 — Week 23 Math: Line Plots with Fractions, Part 1 (5 pts: 3 + 2), with a different tide pool line plot in each set. Remind: each X is one measurement.
12. Slide 11 — Week 23 Math: Fair Shares and Equivalent Fractions, Part 2 (2 pts): share a total equally, stack equivalent fractions, or find how much more is needed.
13. Slide 12 — Week 23 Reading: Figurative Language, Part 1 (5 pts: 3 + 2), with a new part of the story "Aedan's Turn" in each set.
14. Slide 13 — Week 23 Reading: Titles and Dialogue, Part 2 (2 pts): a story title in quotation marks, a speaker-tag comma, or series commas; checked exactly (series with or without the comma before and).
15. Slide 14 — Week 24 Math: Classifying Figures, Part 1 (5 pts: 3 + 2). Tip: triangles get two names, by angles and by sides.
16. Slide 15 — Week 24 Math: Family Tree and Symmetry, Part 2 (2 pts): always, sometimes, or never, or lines of symmetry.
17. Slide 16 — Week 24 Reading: Character Change and Inference, Part 1 (5 pts: 3 + 2), with a short new Sitka story in each set.
18. Slide 17 — Week 24 Reading: Dialogue and Subject-Verb Agreement, Part 2 (2 pts): punctuation inside quotation marks or a compound subject; checked exactly (one item accepts either verb tense).
19. Slide 18 — Putting Skills Together, Part 1 (6 pts, 3 each): a multi-step problem that mixes two weeks and a "which explanation is correct?" item.
20. Slide 19 — Be the Checker, Part 2 (6 pts, 3 each): a math "find the mistake" and a reading/writing "find the mistake." Ask the student to say why each wrong choice is wrong.
21. Slide 20 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, end marks, their/there, your/you're, its/it's). Straight or curly quotes and extra spaces are OK.
22. Slide 21 — Fix the Sentence, Part 2 (4 pts, 2 each): commas in places, dates and a renaming phrase, and find the real subject before choosing the verb; each verb item accepts either verb tense.
23. Slide 22 — Writer's Notebook: Paint the Place: a firsthand paragraph of 3 to 5 sentences about the Unit 5 place the student would most like to explore (practice, XP only), then one scored follow-up item (4 pts). Read the notebook writing aloud together if time allows.
24. Slide 23 — Match-Up Round (10 pts): 5 Unit 5 vocabulary words across math and reading, 2 pts each on the first try.
25. Slide 24 — Looking Back on Unit 5 reflection (bonus XP only).
26. Slide 25 — Unit 5 Boss Challenge (10 pts, 5 each): a multi-step math problem with a different situation in each set (Pittsburgh, Duluth, or Sitka) and a reading/writing item that combines two skills.
27. Slide 26 — Pencil Break (Post): the student completes Page 2 (Checkpoint 5 Challenge) of the same sheet (grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W25.pdf), plus the ★ Challenge if time allows, before seeing the score.
28. Slide 27 — Results (Unit 5 Complete!): the score out of 100 is sent to KingCretot Experience, with a score breakdown by week so you can see what to review. 85 or higher shows "Unit 5: Charting the Course Complete!" with the congratulations message "You did it! Unit 5 is complete!" and a big confetti burst; below 85 shows "Good Work on Checkpoint 5!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

THE COORDINATE PLANE (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: *Quinlan's Trail Map, Revisited Quinlan and his cousin Ramona are back at the ranger station in Yellowstone National Park, plotting points on Ranger Roderick's practice grid. (The grid is made for learning. It is not drawn to scale from a real park map.)* Two steps: Quinlan starts at the footbridge at (8, 2). He walks 3 units left, then 6 units up, then 2 more units left. What ordered pair names the place where he stops? Ordered pair:	**(3, 8)** (also accepted: 3, 8; x = 3, y = 8; x=3,y=8)	Start at (8, 2). Moving left subtracts from x: 8 − 3 = 5, then 5 − 2 = 3. Moving up adds to y: 2 + 6 = 8. He stops at (3, 8).
a2_s0	Set 1: Which point lies ON the x-axis, 4 units from the origin? [(4, 0) / (0, 4) / (4, 4) / (2, 2)]	**(4, 0)**	A point on the x-axis has a y-coordinate of 0. (4, 0) is 4 units right of the origin and 0 units up. (0, 4) is on the y-axis.
a1_s1	Set 2: *Quinlan's Trail Map, Revisited Quinlan and Ramona are back in Yellowstone National Park, marking trailheads and overlooks on Ranger Roderick's practice grid. (The grid is made for learning, not drawn to scale.)* Two steps: Ramona starts at the picnic area at (1, 9). She walks 7 units right, then 5 units down, then 1 more unit right. What ordered pair names the place where she stops? Ordered pair:	**(9, 4)** (also accepted: 9, 4; x = 9, y = 4; x=9,y=4)	Start at (1, 9). Moving right adds to x: 1 + 7 = 8, then 8 + 1 = 9. Moving down subtracts from y: 9 − 5 = 4. She stops at (9, 4).
a2_s1	Set 2: Which point lies ON the y-axis, 6 units above the origin? [(0, 6) / (6, 0) / (6, 6) / (1, 6)]	**(0, 6)**	A point on the y-axis has an x-coordinate of 0. (0, 6) is 0 units right and 6 units up. (6, 0) is on the x-axis.
a1_s2	Set 3: *Quinlan's Trail Map, Revisited Quinlan and Ramona are back at the ranger station in Yellowstone National Park, following Ranger Roderick around the practice grid. (The grid is made for learning, not drawn to scale.)* Two steps: Ranger Roderick starts at the origin. He walks 6 units right, then 9 units up, then 4 units back down. What ordered pair names the place where he stops? Ordered pair:	**(6, 5)** (also accepted: 6, 5; x = 6, y = 5; x=6,y=5)	Start at the origin, (0, 0). Moving right adds to x: 0 + 6 = 6. Moving up adds to y: 0 + 9 = 9, then moving down subtracts: 9 − 4 = 5. He stops at (6, 5).
a2_s2	Set 3: Where is the point (7, 0)? [On the x-axis, 7 units right of the origin / On the y-axis, 7 units above the origin / At the origin / On the y-axis, 0 units above the origin]	**On the x-axis, 7 units right of the origin**	The y-coordinate is 0, so you move 0 units up: the point sits on the x-axis, 7 units right of the origin.

PATTERNS, GRAPHS, AND MAP SCALE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a3_s0	Set 1: Two steps: Rule A: start at 0, add 6. Rule B: start at 0, add 24. Ramona graphs the ordered pairs (Rule A, Rule B). What is the y-coordinate of the point whose x-coordinate is 54? y-coordinate:	**216**	Rule B adds 24 and Rule A adds 6, so every Rule B number is 4 times its Rule A number (24 = 4 × 6). 54 × 4 = 216, so the point is (54, 216).

ID	Question	Answer	Why (explain to student)
a3_s1	Set 2: Two steps: on the practice map, the Visitor Center is at (2, 3) and the Lake Overlook is at (2, 13). Each grid unit stands for 1/4 mile. How many miles apart are they along the straight grid line? Miles:	**2 1/2** (also accepted, with or without the unit word: 5/2, 2.5, 2 and 1/2, 2 2/4)	Both points have an x-coordinate of 2, so subtract the y-coordinates: 13 − 3 = 10 units. Each unit is 1/4 mile: 10 × 1/4 = 10/4 = 2 2/4 = 2 1/2 miles.
a3_s2	Set 3: Two steps: Quinlan's group graphs (hours hiked, miles walked): (1, 3), (3, 9), (5, 15). If the pattern continues, how many miles does the group walk between hour 3 and hour 7? Miles:	**12** (also accepted: 12 miles, 12 mile, 12 mi)	The miles are always 3 times the hours (3 = 3 × 1, 9 = 3 × 3, 15 = 3 × 5). At hour 7 the group has walked 7 × 3 = 21 miles. At hour 3 it had walked 9 miles. 21 − 9 = 12 miles.

TEXT STRUCTURES (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *From Zane's report for the homeschool co-op 1The co-op garden had a problem. By July, the bean plants were drooping, and carrying watering cans from the house took the whole morning. 2To fix this, the students set a rain barrel under the shed's downspout, so rain from the roof would be saved. They also spread straw around each plant to keep the soil from drying out. By August, the beans were climbing the fence again.* Which text structure does this passage MOSTLY use? [Problem and solution / Chronology, because it says By July and By August / Compare and contrast / Description]	**Problem and solution**	Paragraph 1 names a problem (drooping beans and slow watering), and paragraph 2 gives two solutions (the rain barrel and the straw). By July and By August are time words, but one or two time words do not make the whole text chronology. Ask what the whole text is mostly doing.
b2_s0	Set 1: Type your answer: paragraph 2 begins with a three-word signal phrase that points to a SOLUTION. Type those three words. Signal phrase:	**to fix this** (also accepted: to fix this,)	Paragraph 2 begins with the signal phrase To fix this, which points to a solution.
b1_s1	Set 2: *From Yesenia's report for the homeschool co-op 1Yesenia and Zane each built a model home for their Mesa Verde report. Both models are made of brown clay, and both are about the size of a shoebox. However, Zane's model pithouse sits halfway down in a tub of sand, while Yesenia's model cliff dwelling is tucked into a hollow carved in a stack of foam blocks. Unlike the pithouse, her cliff dwelling model has three stories.* Which text structure does Yesenia's paragraph use? [Compare and contrast / Chronology / Cause and effect / Problem and solution]	**Compare and contrast**	The passage tells how the two models are alike (both clay, both shoebox-sized) and how they are different (in sand or in a hollow, one story or three). Both, however, while, and unlike are compare-and-contrast signal words.
b2_s1	Set 2: Type your answer: sentence 3 starts with However. Type the OTHER contrast signal word that appears in the middle of that same sentence. Signal word:	**while**	Sentence 3 starts with However, but the signal word in the MIDDLE of that sentence is while. It shows a difference between the two models.

ID	Question	Answer	Why (explain to student)
b1_s2	Set 3: *From the co-op garden journal 1In June, the co-op planted tomatoes along the north side of the barn. Because the barn's shadow covered that row for most of the day, the plants grew tall and thin, and only a few flowers opened. 2As a result, the students dug up the plants and moved them to the sunny south side. Within three weeks, small green tomatoes began to appear.* Two steps: in paragraph 1, what is the CAUSE, AND what is its EFFECT? [Cause: the barn's shadow blocked the sun. Effect: the plants grew tall and thin with few flowers. / Cause: the plants grew flowers. Effect: the barn cast a shadow. / Cause: the students moved the plants. Effect: it became June. / Cause: green tomatoes appeared. Effect: the barn was built.]	**Cause: the barn's shadow blocked the sun. Effect: the plants grew tall and thin with few flowers.**	Because introduces the cause: the barn's shadow blocked the sun. The effect in paragraph 1 is that the plants grew tall and thin with few flowers. The other choices mix up causes and effects or name things the passage never says.
b2_s2	Set 3: Type your answer: paragraph 2 begins with a three-word signal phrase that shows an EFFECT. Type those three words. Signal phrase:	**as a result** (also accepted: as a result,)	As a result is a cause-and-effect signal phrase. It tells the reader that moving the plants was an effect of the shady row.

SIGNAL-WORD COMMAS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b3_s0	Set 1: Fix this sentence: because the barrel filled up the beans had water all week	**Because the barrel filled up, the beans had water all week.**	Capitalize the first word. When a sentence starts with a because part, put a comma after that whole part, before the main idea.
b3_s1	Set 2: Fix this sentence: the models were finished so zane and yesenia painted them	**The models were finished, so Zane and Yesenia painted them.**	Capitalize the first word and the names Zane and Yesenia. The models were finished and Zane and Yesenia painted them are both complete sentences, so a comma goes before so.
b3_s2	Set 3: Fix this sentence: in the late 1200s the people of mesa verde faced a long drought	**In the late 1200s, the people of Mesa Verde faced a long drought.**	Capitalize the first word and the place name Mesa Verde. Put a comma after the introductory time phrase In the late 1200s.

MEASUREMENT CONVERSIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: *Solveig's Harbor Day, Revisited Solveig is back on the dock in Duluth, Minnesota, helping her Uncle Torvald coil lines and load crates for Captain Teodoro's tugboat.* Two steps: Uncle Torvald's bow line is 7 1/3 yards long. How many INCHES long is it? Inches:	**264** (also accepted: 264 inches, 264 inch, 264 in)	Step 1: 7 yards = 7 × 3 = 21 feet, and 1/3 yard = 1/3 × 3 = 1 foot, so 7 1/3 yards = 22 feet. Step 2: 22 × 12 = 264 inches. (Or: 7 × 36 + 12 = 252 + 12 = 264.)

ID	Question	Answer	Why (explain to student)
c2_s0	Set 1: A crate of propeller parts has a mass of 3.08 kilograms. What is its mass in grams? Grams:	**3,080** (also accepted: 3,080 grams; 3,080 gram; 3,080 g)	Bigger unit to smaller unit, so multiply: 3.08 × 1,000 = 3,080 grams. The decimal point moves 3 places to the right, and a zero fills the empty place.
c1_s1	Set 2: *Solveig's Harbor Day, Revisited Solveig is back on Uncle Torvald's tugboat in the Duluth harbor, measuring tow lines and filling water jugs.* Two steps: a tow line is 3 3/4 yards long. How many INCHES long is it? Inches:	**135** (also accepted: 135 inches, 135 inch, 135 in)	Step 1: 3 yards = 3 × 36 = 108 inches. Step 2: 3/4 yard = 3/4 of 36 = 27 inches. 108 + 27 = 135 inches. (Using 3/4 of a FOOT, 9 inches, is the trap.)
c2_s1	Set 2: A jug holds 4.6 liters of drinking water. How many milliliters is that? Milliliters:	**4,600** (also accepted with a unit word, such as 4,600 milliliters or 4,600 ml, or 2 other unit spellings)	Bigger unit to smaller unit, so multiply: 4.6 × 1,000 = 4,600 milliliters.
c1_s2	Set 3: *Solveig's Harbor Day, Revisited Solveig is back in the tugboat's tiny galley in the Duluth harbor, helping Captain Teodoro's cook measure soup and cable.* Two steps: the galley cook makes 6 3/4 gallons of soup. How many cups of soup is that? Cups:	**108** (also accepted: 108 cups, 108 cup, 108 c)	Step 1: 1 gallon = 16 cups, so 6 gallons = 6 × 16 = 96 cups. Step 2: 3/4 gallon = 3/4 of 16 = 12 cups. 96 + 12 = 108 cups.
c2_s2	Set 3: A cable is 2.07 meters long. How many centimeters long is it? Centimeters:	**207** (also accepted: 207 centimeters, 207 centimeter, 207 cm)	Bigger unit to smaller unit, so multiply: 2.07 × 100 = 207 centimeters. The decimal point moves 2 places to the right.

SMALLER UNITS TO BIGGER UNITS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c3_s0	Set 1: Two steps: a hatch cover on the dock is 143 inches wide. Write the width in feet and inches, like 5 feet 2 inches. Feet and inches:	**11 feet 11 inches** (also accepted: 11 ft 11 in; 11 feet, 11 inches; 11 feet and 11 inches; 11 ft, 11 in; 11 ft. 11 in; 11 foot 11 inches; 11 feet 11 inch)	143 ÷ 12 = 11 R11, because 11 × 12 = 132 and 143 − 132 = 11. So 143 inches = 11 feet 11 inches. Check: 132 + 11 = 143.
c3_s1	Set 2: Two steps: a bag of washers weighs 92 ounces. Write the weight in pounds and ounces, like 2 pounds 3 ounces. Pounds and ounces:	**5 pounds 12 ounces** (also accepted: 5 lb 12 oz; 5 pounds, 12 ounces; 5 pounds and 12 ounces; 5 lb, 12 oz; 5 pound 12 ounces; 5 pounds 12 ounce; 5 3/4 pounds; 5 3/4 pound; 5 3/4 lb; 5.75 pounds; 5.75 lb)	92 ÷ 16 = 5 R12, because 5 × 16 = 80 and 92 − 80 = 12. So 92 ounces = 5 pounds 12 ounces. (12 ounces is 12/16 = 3/4 pound, so that is also 5 3/4 pounds.)
c3_s2	Set 3: Two steps: the cook has 13 pints of fish chowder. How many quarts is that? Write a mixed number. Quarts:	**6 1/2** (also accepted, with or without the unit word: 13/2, 6.5, 6 and 1/2, 6 2/4)	2 pints make 1 quart, so divide: 13 ÷ 2 = 6 R1. The leftover pint is half of a quart, so 13 pints = 6 1/2 quarts.

FIRSTHAND AND SECONDHAND ACCOUNTS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
h1_s0	Set 1: *Two accounts of the Eastside Rocket Club launch 1From Achille's science notebook: I held my breath while our club leader counted down. Our rocket hissed, shot straight up, and left a smell like burnt matches. My hands were still shaking when I found its orange parachute in the soccer field. 2From the club newsletter, written the next week by a parent who missed the launch: On Saturday, the Eastside Rocket Club launched 12 model rockets. According to the club's records, the highest flight reached about 300 feet. Members said the parachutes opened on 11 of the 12 rockets.* Which clue BEST shows that Account 1 is a firsthand account? [Achille uses I, my, and our, and he tells what he smelled and felt at the launch. / It tells exactly how many rockets were launched. / It was printed in the club newsletter. / It uses the words according to the club's records.]	**Achille uses I, my, and our, and he tells what he smelled and felt at the launch.**	Achille uses I, my, and our, and he tells what he smelled and felt with his own body. Only someone who was there could write that. The rocket count and the newsletter come from Account 2, and according to the records shows a secondhand source.
h2_s0	Set 1: Type your answer: in Account 2, which TWO words begin the phrase that shows the writer got facts from another source? (___ ___ the club's records) Two words:	**according to**	According to the club's records shows that the newsletter writer learned the facts from another source. That is a clue to a secondhand account.
h1_s1	Set 2: *Two accounts of the same stormy night 1From Abena's diary, Tuesday night: The lights flickered twice, and then everything went dark. My cousin Achille and I ate cold pizza by flashlight, and we could hear the wind rattling the back door. I was a little scared, but mostly I felt like we were camping in our own kitchen. 2From the town newspaper, Wednesday: A storm knocked out power to about 4,000 homes on Tuesday night. The power company said its crews had turned the power back on in most homes by 6 a.m.* Which question can ONLY Account 2 answer? [About how many homes lost power? / What did Abena and Achille eat for dinner? / What could the cousins hear during the storm? / How did Abena feel when the lights went out?]	**About how many homes lost power?**	Only the newspaper gives the big-picture number of homes (about 4,000). What the cousins ate, what they heard, and how Abena felt are close-up details that only her firsthand diary gives.
h2_s1	Set 2: Type your answer: in Account 2, the facts about turning the power back on came from a source. Type the word that completes it: the power ___. Word:	**company** (also accepted: power company, the power company)	The newspaper writer says the power company gave the facts about the repairs. A secondhand account depends on sources like this.

ID	Question	Answer	Why (explain to student)
h1_s2	Set 3: *From an interview with Ms. Galloway, a school bus driver 1"I was driving my afternoon route when the hail started. It sounded like a thousand marbles bouncing on the roof. I pulled over by the library, and the children and I waited until it stopped. Later that night, I heard on the radio that the hail had cracked windows all across town."* Two steps: which explanation of Ms. Galloway's account is correct? [It is firsthand about the hail on her bus, but secondhand about the windows across town. / It is firsthand about everything, because she lives in the town. / It is secondhand about everything, because it is an interview. / It is firsthand about the windows, because she heard about them on the radio.]	**It is firsthand about the hail on her bus, but secondhand about the windows across town.**	Firsthand about WHAT? Ms. Galloway was there for the hail on her bus, so that part is firsthand. She did not see the windows across town; she heard about them on the radio, so that part is secondhand.
h2_s2	Set 3: Type your answer: where did Ms. Galloway learn about the cracked windows across town? She heard it on the ___. Word:	**radio** (also accepted: the radio)	She says, "I heard on the radio that the hail had cracked windows all across town." The radio was her source for that part.

DATES, QUOTATIONS, AND PRONOUNS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
h3_s0	Set 1: Fix this sentence: On May 3 2026 the club launched twelve rockets.	**On May 3, 2026, the club launched twelve rockets.**	Put a comma between the day and the year, and another comma after the year because the sentence keeps going.
h3_s1	Set 2: Fix this sentence: Abena wrote "The lights went out at nine".	**Abena wrote, "The lights went out at nine."**	Put a comma after the speaker tag Abena wrote. The period at the end of the quotation goes inside the closing quotation mark.
h3_s2	Set 3: Fix this sentence: Achille and me watched the launch from the bleachers.	**Achille and I watched the launch from the bleachers.**	Achille and I is the subject of the sentence, so use I, not me. Test it: you would say I watched, not me watched.

LINE PLOTS WITH FRACTIONS (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: *Vikram's Tide Pool Survey, Revisited Vikram and his cousin Uzoma are back at the tide pools around Haystack Rock in Cannon Beach, Oregon. Mr. Varga plotted the lengths of the periwinkle snails they measured, without picking any up. [Line plot "Periwinkle Snail Lengths (inches)"; X's: 3/8 (1); 1/2 (3); 5/8 (2); 3/4 (3); 1 (2).] Each X is one snail. Each tick mark is 1/8 inch.* Two steps: what is the total length of ALL the snails that measured 3/4 inch, together with ALL the snails that measured 5/8 inch? Inches:	**3 1/2** (also accepted, with or without the unit word: 7/2, 3.5, 3 2/4, 3 4/8, 28/8, 14/4, 3 and 1/2)	Three snails at 3/4 inch: 3 × 3/4 = 9/4 = 18/8. Two snails at 5/8 inch: 2 × 5/8 = 10/8. Total: 18/8 + 10/8 = 28/8 = 3 4/8 = 3 1/2 inches.

ID	Question	Answer	Why (explain to student)
d2_s0	Set 1: How much longer is the longest snail than the shortest snail? [5/8 inch / 1 3/8 inches / 3/8 inch / 7/8 inch]	**5/8 inch**	Longest: 1 inch = 8/8. Shortest: 3/8. 8/8 − 3/8 = 5/8 inch. (1 3/8 comes from adding instead of subtracting.)
d1_s1	Set 2: *Vikram's Tide Pool Survey, Revisited Vikram and Uzoma are back at the tide pools around Haystack Rock in Cannon Beach, Oregon. Ms. Uchida measured the shell widths of purple shore crabs to the nearest 1/8 inch, and Mr. Varga plotted them. [Line plot "Purple Shore Crab Widths (inches)"; X's: 1 1/4 (2); 1 3/8 (1); 1 1/2 (4); 1 3/4 (2); 1 7/8 (1).] Each X is one crab. Each tick mark is 1/8 inch.* Two steps: how much more is the total width of ALL the 1 1/2-inch crabs than the total width of ALL the 1 3/4-inch crabs? Inches:	**2 1/2** (also accepted, with or without the unit word: 5/2, 2.5, 2 2/4, 2 4/8, 20/8, 10/4, 2 and 1/2)	Four crabs at 1 1/2 inches: 4 × 1 1/2 = 6 inches. Two crabs at 1 3/4 inches: 2 × 1 3/4 = 3 1/2 inches. Difference: 6 − 3 1/2 = 2 1/2 inches.
d2_s1	Set 2: How many crabs are AT LEAST 1 1/2 inches wide? [7 crabs / 3 crabs / 4 crabs / 10 crabs]	**7 crabs**	At least 1 1/2 inches means 1 1/2 or wider: 4 crabs at 1 1/2, 2 at 1 3/4, and 1 at 1 7/8. 4 + 2 + 1 = 7 crabs. The crabs at exactly 1 1/2 count.
d1_s2	Set 3: *Vikram's Tide Pool Survey, Revisited Vikram and Uzoma are back at the tide pools around Haystack Rock in Cannon Beach, Oregon. Vikram measured one arm on each of 9 ochre sea stars to the nearest 1/4 inch, without touching them, and plotted the lengths. [Line plot "Ochre Sea Star Arm Lengths (inches)"; X's: 2 1/4 (1); 2 1/2 (2); 2 3/4 (3); 3 (1); 3 1/2 (2).] Each X is one sea star arm. Each tick mark is 1/4 inch.* Two steps: what is the total length of ALL the 2 3/4-inch arms, together with ALL the 3 1/2-inch arms? Inches:	**15 1/4** (also accepted, with or without the unit word: 61/4, 15.25, 15 2/8, 15 and 1/4)	Three arms at 2 3/4 inches: 3 × 2 3/4 = 6 9/4 = 8 1/4 inches. Two arms at 3 1/2 inches: 2 × 3 1/2 = 7 inches. Total: 8 1/4 + 7 = 15 1/4 inches.
d2_s2	Set 3: What is the difference between the longest arm and the shortest arm? [1 1/4 inches / 5 3/4 inches / 1 1/2 inches / 1 inch]	**1 1/4 inches**	Longest: 3 1/2 = 3 2/4. Shortest: 2 1/4. 3 2/4 − 2 1/4 = 1 1/4 inches. (5 3/4 comes from adding.)

FAIR SHARES AND EQUIVALENT FRACTIONS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d3_s0	Set 1: Two steps: Ms. Uchida's 4 sample jars hold 3/8 cup, 1/2 cup, 3/4 cup, and 1 3/8 cups of seawater. If all the water were shared equally, how much would each jar hold? Cups:	**3/4** (also accepted, with or without the unit word: 6/8, 0.75, 12/16)	Total in eighths: 3/8 + 4/8 + 6/8 + 11/8 = 24/8 = 3 cups. Share among 4 jars: 3 ÷ 4 = 3/4 cup in each jar.
d3_s1	Set 2: Uzoma's list of crab widths, in inches: 1 1/2, 1 3/8, 1 6/8, 1 4/8, 1 3/4, 2, 1 2/4, 1 1/4. When he makes a line plot, how many X's go above 1 1/2? X's above 1 1/2:	**3** (also accepted with a unit word, such as 3 x's or 3 marks, or 2 other unit spellings)	Equivalent fractions share a stack. 1 4/8 = 1 1/2 and 1 2/4 = 1 1/2, so the 1 1/2 stack gets 1 1/2, 1 4/8, and 1 2/4: 3 X's. (1 6/8 goes with 1 3/4 instead.)
d3_s2	Set 3: Three steps: Mr. Varga needs 5 cups of seawater. His 6 jars hold 1/2, 5/8, 3/4, 3/4, 7/8, and 1 cup. How many more cups does he need? Cups:	**1/2** (also accepted, with or without the unit word: 2/4, 4/8, 0.5, 8/16)	Total in eighths: 4/8 + 5/8 + 6/8 + 6/8 + 7/8 + 8/8 = 36/8 = 4 4/8 = 4 1/2 cups. Still needed: 5 − 4 1/2 = 1/2 cup.

FIGURATIVE LANGUAGE (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: *From "Aedan's Turn," a new story from the Nashville fiddle camp 1The afternoon before the camp jam, Aedan sat on the cabin steps with his banjo, as still as a statue. 2"My hands are blocks of ice," he told Adisa. 3The wind tugged at the pages of his music, trying to steal them away.* Two steps: sentence 1 says Aedan sat "as still as a statue." Which answer names the kind AND tells what it shows? [A simile; it shows he is frozen with nerves. / A metaphor; it shows he is frozen with nerves. / A simile; it shows he is a real statue in the camp. / Personification; it shows he is excited to play.]	**A simile; it shows he is frozen with nerves.**	As still as a statue compares Aedan to a statue using as, so it is a simile. He is not really a statue; the comparison shows that nerves have frozen him in place.
e2_s0	Set 1: Type your answer: in sentence 3, the wind "tugged" at the pages, "trying to steal them away." What kind of figurative language is this? Kind:	**personification** (also accepted: personify, personifying)	Wind cannot really tug or try to steal anything. Giving the wind human actions is personification.
e1_s1	Set 2: *From "Aedan's Turn," a new story from the Nashville fiddle camp 1That evening, the jam circle was a beehive. 2Fiddles buzzed, guitars hummed, and Mr. Aldo's boot kept time on the wooden porch. 3When it was Aedan's turn, he took a deep breath and jumped in with both feet. 4The crowd clapped along to every note.* Two steps: "the jam circle was a beehive." Which answer names the kind AND tells what it shows? [A metaphor; the circle is crowded, busy, and buzzing with music. / A simile; the circle is crowded, busy, and buzzing with music. / A metaphor; real bees are flying around the campers. / Hyperbole; the circle is quiet and empty.]	**A metaphor; the circle is crowded, busy, and buzzing with music.**	The sentence says the circle WAS a beehive, with no like or as, so it is a metaphor. A beehive is crowded and full of buzzing, so the circle is busy and full of music.
e2_s1	Set 2: Type your answer: sentence 3 uses an idiom that means "started boldly, all at once." He jumped in with ___ ___. Type the two missing words. Two words:	**both feet**	Jumped in with both feet is an idiom. Aedan did not really jump; he started playing boldly, all at once.
e1_s2	Set 3: *From "Aedan's Turn," a new story from the Nashville fiddle camp 1After the jam, the campers drifted down to the river. 2The water whispered against the rocks, and fireflies blinked like tiny lanterns. 3Aedan hummed softly, his banjo resting across his knees.* Which words BEST describe the MOOD of this passage? [Peaceful and calm / Tense and worried / Angry and loud / Gloomy and lonely]	**Peaceful and calm**	Drifted, whispered, tiny lanterns, hummed softly, and resting all create a calm, quiet feeling. Nothing in the passage is tense, angry, or gloomy.
e2_s2	Set 3: Type your answer: in sentence 2, fireflies "blinked like tiny lanterns." What kind of figurative language is this? Kind:	**simile** (also accepted with "a" or "an" or in the plural)	Fireflies blinked like tiny lanterns compares two unlike things using like, so it is a simile.

TITLES AND DIALOGUE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Fix this sentence: In the story Aedan's Turn, the banjo player is nervous.	**In the story "Aedan's Turn," the banjo player is nervous.**	The title of a story goes inside quotation marks, and the comma after the title goes inside the closing quotation mark.
e3_s1	Set 2: Fix this sentence: "Just play" Adisa said.	**"Just play," Adisa said.**	When the speaker tag comes after the spoken words, a comma goes inside the closing quotation mark.
e3_s2	Set 3: Fix this sentence: The fiddles buzzed the guitars hummed and the crowd clapped.	**The fiddles buzzed, the guitars hummed, and the crowd clapped.** (also accepted: The fiddles buzzed, the guitars hummed and the crowd clapped.)	Use commas to separate the three parts in a series: the fiddles buzzed, the guitars hummed, and the crowd clapped.

CLASSIFYING FIGURES (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
f1_s0	Set 1: *Ximena's Design Studio, Revisited Ximena is back at the bridge design studio in Pittsburgh, Pennsylvania, checking Aunt Winnie's truss drawings.* Two steps: an isosceles triangle in a truss has a perimeter of 41 feet. Each of its two equal sides is 13 feet long. How long is the third side? Feet:	**15** (also accepted: 15 feet, 15 foot, 15 ft)	Step 1: the two equal sides together are 13 + 13 = 26 feet. Step 2: the third side is 41 − 26 = 15 feet. Check: 13 + 13 + 15 = 41.
f2_s0	Set 1: Type your answer: a steel plate is a triangle with angles of 20°, 60°, and 100°, and no equal sides. Give its full name, angles first (two words, like "acute isosceles"). Full name:	**obtuse scalene** (also accepted: obtuse scalene triangle, scalene obtuse, scalene obtuse triangle; with "a" or "an" or in the plural)	The 100° angle is more than 90°, so the triangle is obtuse. No sides are equal, so it is scalene. Its full name is an obtuse scalene triangle.
f1_s1	Set 2: *Ximena's Design Studio, Revisited Ximena is back at the studio in Pittsburgh, Pennsylvania, helping Ms. Whitaker check the deck panels.* Two steps: a rectangle-shaped deck has a perimeter of 58 feet. One side is 17 feet long. How long is a side next to it? Feet:	**12** (also accepted: 12 feet, 12 foot, 12 ft)	Opposite sides of a rectangle are equal. Step 1: the two 17-foot sides use 2 × 17 = 34 feet. Step 2: the other two sides share 58 − 34 = 24 feet, so each one is 24 ÷ 2 = 12 feet. Check: 17 + 12 + 17 + 12 = 58.
f2_s1	Set 2: Type your answer: riddle. I have 4 sides and exactly 1 pair of parallel sides, and my two slanted sides are equal. What am I? (two words) Shape:	**isosceles trapezoid** (also accepted with "a" or "an" or in the plural)	Exactly 1 pair of parallel sides makes it a trapezoid. Two equal non-parallel (slanted) sides make it an isosceles trapezoid.

ID	Question	Answer	Why (explain to student)
f1_s2	Set 3: *Ximena's Design Studio, Revisited Ximena is back at the studio in Pittsburgh, Pennsylvania, watching Mr. Wojcik cut steel plates for a scale model.* Two steps: Mr. Wojcik cuts 4 separate equilateral triangles and 2 separate squares. Every side of every plate is 7 inches long. How many inches of edge do the 6 plates have in all? Inches:	**140** (also accepted: 140 inches, 140 inch, 140 in)	Step 1: one equilateral triangle has 3 sides: 3 × 7 = 21 inches, and 4 triangles use 4 × 21 = 84 inches. Step 2: one square has 4 sides: 4 × 7 = 28 inches, and 2 squares use 2 × 28 = 56 inches. Total: 84 + 56 = 140 inches.
f2_s2	Set 3: Type your answer: a steel plate is shaped like a regular octagon (8 equal sides and 8 equal angles). How many lines of symmetry does it have? Lines:	**8** (also accepted: 8 lines, 8 line, eight)	A regular polygon has as many lines of symmetry as it has sides. A regular octagon has 8 sides, so it has 8 lines of symmetry.

FAMILY TREE AND SYMMETRY (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
f3_s0	Set 1: Always, sometimes, or never: a trapezoid is a parallelogram. (Use the KCE Pathways definition: a trapezoid has at least 1 pair of parallel sides.) [Sometimes, when it has 2 pairs of parallel sides / Always / Never / Sometimes, when it has one right angle]	**Sometimes, when it has 2 pairs of parallel sides**	In KCE Pathways a trapezoid has AT LEAST one pair of parallel sides. A trapezoid with 2 pairs is also a parallelogram, but one with exactly 1 pair is not, so the answer is sometimes.
f3_s1	Set 2: How many lines of symmetry does a rhombus that is NOT a square have? [2, along its diagonals / 4, like a square / 0, because its sides slant / 1, straight down the middle]	**2, along its diagonals**	A rhombus folds onto itself along both diagonals, so it has 2 lines of symmetry. If it is not a square, a fold straight across the middle does not match.
f3_s2	Set 3: Which statement is NEVER true? [A triangle has a pair of parallel sides. / A rectangle is a square. / A trapezoid has two right angles. / An isosceles triangle is acute.]	**A triangle has a pair of parallel sides.**	Any two sides of a triangle meet at a corner, so they can never be parallel. The other three can be true for some shapes: a square is a rectangle, a right trapezoid has 2 right angles, and many isosceles triangles are acute.

CHARACTER CHANGE AND INFERENCE (2, 3 POINTS)

ID	Question	Answer	Why (explain to student)
g1_s0	Set 1: *From "Arvid and the Knot," a new story from Sitka, Alaska 1Arvid would not step onto Grandpa Arne's boat. "It wobbles," he said, gripping the dock post with both hands. 2Anya didn't laugh. She sat on the edge of the dock and showed him the knot Grandpa had taught her, the one that held the boat snug against the dock. 3Arvid tied it once, then again, then tugged it as hard as he could. It held. 4The next morning, Arvid was the one who untied the line and hopped aboard first.* Which pair of traits BEST shows how Arvid changes from the beginning to the end? [From fearful to confident / From confident to fearful / From kind to selfish / He stays exactly the same.]	**From fearful to confident**	At first Arvid grips the dock post and will not get on the boat, so he is fearful. At the end he unties the line and hops aboard first, so he is confident. The other choices reverse the change or name traits the story never shows.

ID	Question	Answer	Why (explain to student)
g2_s0	Set 1: Type your answer: at the turning point in paragraph 3, Arvid tugs the knot as hard as he can. Type the ONE word that tells what the knot did. Word:	**held** (also accepted: it held)	Paragraph 3 ends, “It held.” The knot holding is the turning point: now Arvid trusts that the boat will stay put.
g1_s1	Set 2: *From “Amina’s Visit,” a new story from Sitka, Alaska 1On her first morning in Sitka, Amina kept checking her phone. “No signal out here?” she groaned as Grandpa Arne’s boat left the harbor. 2Anya didn’t argue. She just handed Amina a pair of binoculars. 3An hour later, Amina was leaning over the rail, pointing at a bald eagle circling above the water. “Look, Anya, look!” she whispered. Her phone stayed in her pocket for the rest of the trip.* What can you INFER about Amina from paragraph 3? [She has become interested in what she sees and has forgotten about her phone. / She is still bored and wants to go back to the harbor. / She is afraid of the bald eagle. / She is angry at Anya for taking her phone.]	**She has become interested in what she sees and has forgotten about her phone.**	Amina leans over the rail, points, and whispers “Look, Anya, look!” and her phone stays in her pocket. Those clues, plus what you know, show she is now interested and has forgotten her phone. Nothing in the text supports the other choices.
g2_s1	Set 2: Type your answer: what does Anya hand Amina in paragraph 2, the gift that starts Amina’s change? (one word) Object:	**binoculars** (also accepted: binocular, the binoculars, pair of binoculars; with “a” or “an” or in the plural)	Anya hands Amina a pair of binoculars in paragraph 2. Looking through them is what starts Amina’s change.
g1_s2	Set 3: *From “Mending Day,” a new story from Sitka, Alaska 1Grandpa Arne’s fingers moved slowly as he tried to mend the torn net. He sighed. “These old hands are not as quick as they used to be.” 2Arvid sat down beside him. “Then you be the teacher, and I’ll be the hands.” 3By sunset, the net was mended, and Grandpa was laughing at Arvid’s lopsided knots.* Which THEME does this passage BEST show? [When people work together, they can do what one person could not do alone. / Fishing nets can tear. / Knots are the hardest part of fishing. / Always tie your knots tightly.]	**When people work together, they can do what one person could not do alone.**	Grandpa knows how to mend the net, and Arvid has quick hands. Together they finish the job by sunset. That shows a full-sentence message about life. The other choices are a fact, an opinion, or a command.
g2_s2	Set 3: Type your answer: the message about life that a story shows, often through how a character changes, is called its ___. Word:	**theme** (also accepted: the theme; with “a” or “an” or in the plural)	The message about life a story shows, often through how a character changes, is its theme.

DIALOGUE AND SUBJECT-VERB AGREEMENT (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
g3_s0	Set 1: Fix this sentence: "It wobbles" Arvid said.	**"It wobbles," Arvid said.**	When the speaker tag comes after the spoken words, a comma goes inside the closing quotation mark.
g3_s1	Set 2: Fix this sentence: Arvid and Anya ties the boat to the dock.	**Arvid and Anya tie the boat to the dock.** (also accepted: Arvid and Anya tied the boat to the dock.)	Arvid and Anya is a compound subject (two people), so it needs the plural verb tie, not ties.

ID	Question	Answer	Why (explain to student)
g3_s2	Set 3: Fix this sentence: Amina asked, "do they always fly that low"?	**Amina asked, "Do they always fly that low?"**	The first word of a quoted sentence is capitalized, so Do needs a capital D. The question mark belongs to Amina's words, so it goes inside the closing quotation mark.

PUTTING SKILLS TOGETHER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Three steps: on Ranger Roderick's practice map, the footbridge is at (2, 6) and the bench is at (9, 6). Each grid unit stands for 20 yards. How many FEET apart are they along the grid line? Feet:	**420** (also accepted: 420 feet, 420 foot, 420 ft)	Step 1: both points have a y-coordinate of 6, so subtract the x-coordinates: 9 − 2 = 7 units. Step 2: 7 × 20 = 140 yards. Step 3: 1 yard = 3 feet, so 140 × 3 = 420 feet.
x2_s0	Set 1: Which explanation is correct? How many ounces are in 2 3/8 pounds? [38 ounces, because 2 × 16 = 32 and 3/8 of 16 is 6 / 35 ounces, because 2 × 16 = 32 and you add the 3 / 19 ounces, because 2 3/8 × 8 = 19 / 40 ounces, because you round 2 3/8 up to 2 1/2]	**38 ounces, because 2 × 16 = 32 and 3/8 of 16 is 6**	1 pound = 16 ounces. 2 pounds = 32 ounces, and 3/8 of 16 = 6 ounces. 32 + 6 = 38 ounces. (35 treats 3/8 as 3 ounces, and 19 multiplies by 8 instead of 16.)
x1_s1	Set 2: Two steps: a square railing panel in Ximena's design has a perimeter of 9 feet. How many INCHES long is each side? Inches:	**27** (also accepted: 27 inches, 27 inch, 27 in)	Step 1: a square has 4 equal sides, so each side is 9 ÷ 4 = 2 1/4 feet. Step 2: 2 feet = 24 inches, and 1/4 foot = 3 inches. 24 + 3 = 27 inches. Check: 4 × 27 = 108 inches = 9 feet.
x2_s1	Set 2: Which explanation is correct for changing 0.45 kilogram into grams? [450 grams, because 1 kilogram = 1,000 grams, so multiply 0.45 × 1,000 / 45 grams, because you multiply by 100 / 4.5 grams, because you move the decimal point one place / 4,500 grams, because you move the decimal point 4 places]	**450 grams, because 1 kilogram = 1,000 grams, so multiply 0.45 × 1,000**	Kilo- means 1,000, so a bigger unit (kilograms) becomes a smaller unit (grams) by multiplying by 1,000. The decimal point moves 3 places to the right: 0.45 × 1,000 = 450 grams.
x1_s2	Set 3: Two steps: Vikram's 4 jars hold 3/4, 1/2, 7/8, and 7/8 cup of seawater. He pours them all into one bottle. How many PINTS of seawater is that? (1 pint = 2 cups) Pints:	**1 1/2** (also accepted, with or without the unit word: 3/2, 1.5, 1 2/4, 1 and 1/2)	Step 1: total in eighths: 6/8 + 4/8 + 7/8 + 7/8 = 24/8 = 3 cups. Step 2: 2 cups make 1 pint, so 3 ÷ 2 = 1 1/2 pints.
x2_s2	Set 3: Which explanation is correct? Why is every equilateral triangle also an isosceles triangle? [An isosceles triangle has at least 2 equal sides, and an equilateral triangle has 3 equal sides. / Every triangle is isosceles. / An equilateral triangle always has a right angle. / Isosceles means no equal sides.]	**An isosceles triangle has at least 2 equal sides, and an equilateral triangle has 3 equal sides.**	An isosceles triangle has AT LEAST two equal sides. An equilateral triangle has three equal sides, so it also has at least two: it is a special isosceles triangle.

BE THE CHECKER (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x3_s0	Set 1: Find the mistake: Isadora changes 18 feet into yards and writes 54 yards. What went wrong? [She multiplied by 3. Smaller unit to bigger unit means divide: 18 ÷ 3 = 6 yards. / Nothing. 18 feet = 54 yards. / She should multiply by 12: 18 × 12 = 216 yards. / She should divide by 12: 18 ÷ 12 = 1 1/2 yards.]	**She multiplied by 3. Smaller unit to bigger unit means divide: 18 ÷ 3 = 6 yards.**	Feet are smaller than yards, so changing feet into yards means DIVIDING by 3: 18 ÷ 3 = 6 yards. Isadora multiplied, which gives a bigger number of a bigger unit.
x4_s0	Set 1: Find the mistake: Ezekiel says, "'My backpack is a boulder' is a simile, because it compares two things." What is wrong? [It is a metaphor. It says the backpack IS a boulder, with no like or as. / Nothing. Any comparison is a simile. / It is an idiom, because people say it every day. / It is literal. The backpack really is a boulder.]	**It is a metaphor. It says the backpack IS a boulder, with no like or as.**	A simile must use like or as. "My backpack is a boulder" says one thing IS another with no like or as, so it is a metaphor. (It shows the backpack feels very heavy.)
x3_s1	Set 2: Find the mistake: Philippa finds how much longer a 1 1/4-inch crab is than a 5/8-inch crab. She writes 1 2/8 − 5/8 = 1 3/8. What should she fix? [Regroup: 1 2/8 = 10/8, and 10/8 − 5/8 = 5/8 inch. / Nothing. The answer is 1 3/8 inches. / Add instead: 1 2/8 + 5/8 = 1 7/8 inches. / Subtract the denominators: 8 − 8 = 0.]	**Regroup: 1 2/8 = 10/8, and 10/8 − 5/8 = 5/8 inch.**	Rename 1 1/4 as 1 2/8. You cannot take 5/8 from 2/8, so regroup: 1 2/8 = 10/8. 10/8 − 5/8 = 5/8 inch. Philippa subtracted 2/8 from 5/8 instead, which flips the problem.
x4_s1	Set 2: Find the mistake: Tobiah says, "A biography of a sea captain, written 100 years after the captain died, is a firsthand account because it has names and dates." What is wrong? [It is secondhand. The biographer was not there; names and dates do not make an account firsthand. / Nothing. Any account with dates is firsthand. / It is firsthand because biographies use the word I. / It is not an account at all, because it is a book.]	**It is secondhand. The biographer was not there; names and dates do not make an account firsthand.**	Firsthand means the writer was there. A writer working 100 years later was not there, so a biography is secondhand, even with real names and dates.
x3_s2	Set 3: Find the mistake: Beatrix says a triangle with angles of 90°, 60°, and 30° is an acute triangle, because two of its angles are acute. What should she fix? [One right angle makes it a right triangle. / Nothing. It is an acute triangle. / It is obtuse, because 90° is a big angle. / It is equilateral, because the angles add to 180°.]	**One right angle makes it a right triangle.**	A triangle with a 90° angle is a right triangle. The other two angles are acute, but one right angle decides the name.
x4_s2	Set 3: Read: Ottilie slammed her math book shut. Then she opened it again and worked until dinner. Lazlo infers, "Ottilie gives up easily." What is wrong? [He ignored the second clue. She opened the book again and kept working, so she is frustrated but determined. / Nothing. Slamming a book proves she gives up. / An inference should never use clues from the text. / Ottilie is a math teacher, so she cannot give up.]	**He ignored the second clue. She opened the book again and kept working, so she is frustrated but determined.**	A strong inference uses ALL the clues. Ottilie slams the book (frustrated), but she opens it again and works until dinner, so she is determined, not someone who gives up.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: Captain Teodoro said, "the bridge is going up".	**Captain Teodoro said, "The bridge is going up."**	The first word of a quoted sentence is capitalized, so The needs a capital T. The period goes inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The tugboat crew ate there lunch in the galley.	**The tugboat crew ate their lunch in the galley.**	Their shows ownership: the lunch belongs to the crew. There tells where something is.
s3_s0	Set 1: Fix this sentence: Vikram visited Cannon Beach Oregon in June.	**Vikram visited Cannon Beach, Oregon, in June.**	Put a comma between the city and the state, and another comma after the state because the sentence keeps going.
s4_s0	Set 1: Fix this sentence: The crabs in the tide pool hides under the rocks.	**The crabs in the tide pool hide under the rocks.** (also accepted: The crabs in the tide pool hid under the rocks.)	The subject is crabs (plural), not pool, so use hide, not hides. The words “in the tide pool” do not change the subject.
s1_s1	Set 2: Fix this sentence: Grandpa Arne asked "Are you ready to go"?	**Grandpa Arne asked, "Are you ready to go?"**	Put a comma after the speaker tag Grandpa Arne asked. The question mark belongs to his words, so it goes inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: Your going to need a sharper pencil for the line plot.	**You're going to need a sharper pencil for the line plot.**	You're is short for you are: you are going to need a sharper pencil. Your shows ownership, as in your pencil.
s3_s1	Set 2: Fix this sentence: Aunt Winnie a structural engineer checked the drawing.	**Aunt Winnie, a structural engineer, checked the drawing.**	A structural engineer renames Aunt Winnie in the middle of the sentence, so it gets a comma on each side.
s4_s1	Set 2: Fix this sentence: Each of the campers carry a fiddle case.	**Each of the campers carries a fiddle case.** (also accepted: Each of the campers carried a fiddle case.)	The subject is Each, which means each ONE, so it is singular and needs carries, not carry. The words “of the campers” do not change the subject.
s1_s2	Set 3: Fix this sentence (Anya’s words are an exclamation): "Look at the whale"! Anya shouted.	**"Look at the whale!" Anya shouted.**	The exclamation point belongs to Anya’s words, so it goes inside the closing quotation mark, before the speaker tag.
s2_s2	Set 3: Fix this sentence: Its time to plot the last point.	**It's time to plot the last point.**	It's means it is: it is time to plot the last point. Use the apostrophe.
s3_s2	Set 3: Fix this sentence: The club met in Duluth Minnesota on March 4 2026.	**The club met in Duluth, Minnesota, on March 4, 2026.**	Put commas between the city and the state and after the state. Put a comma between the day and the year.
s4_s2	Set 3: Fix this sentence: The bridges over the river is painted gold.	**The bridges over the river are painted gold.** (also accepted: The bridges over the river were painted gold.)	The subject is bridges (plural), not river, so use are, not is. The words “over the river” do not change the subject.

WRITER'S NOTEBOOK: PAINT THE PLACE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Which sentence uses a comma after an introductory word AND a simile? [Finally, the fog rolled away like a curtain opening. / Finally the fog rolled away like a curtain opening. / Finally, the fog was a curtain. / Finally, the fog rolled away.]	**Finally, the fog rolled away like a curtain opening.**	It has a comma after the introductory word Finally, and like a curtain opening is a simile. One choice is missing the comma, one is a metaphor (no like or as), and one has no figurative language.
k1_s1	Set 2: Which sentence punctuates dialogue correctly AND uses personification? ["Listen," Grandpa said, "the harbor is waking up." / "Listen" Grandpa said "the harbor is waking up." / "Listen," Grandpa said, "the harbor is quiet." / Listen, Grandpa said, the harbor is waking up.]	**"Listen," Grandpa said, "the harbor is waking up."**	Commas set off the speaker tag inside the quotation, and a harbor cannot really wake up, so that is personification. The others are missing commas, have no personification, or have no quotation marks.
k1_s2	Set 3: Which sentence gives a FIRSTHAND detail AND uses commas in a series correctly? [I saw puffins, sea stars, and a crab from the beach. / I saw puffins sea stars, and a crab from the beach. / Visitors can see puffins, sea stars, and crabs from the beach. / I saw, puffins, sea stars and, a crab from the beach.]	**I saw puffins, sea stars, and a crab from the beach.**	It uses I, so it is a firsthand detail, and it puts a comma after each item in the series except the last. One choice is missing a comma, one is not firsthand (no I), and one puts commas in the wrong places.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match ordered pair	**two numbers in a set order that name a point**	2 points if matched on the first try.
match_s0_2	Set 1: Match mixed measurement	**a measurement that uses two units at once**	2 points if matched on the first try.
match_s0_3	Set 1: Match line plot	**a number line with an X for each piece of data**	2 points if matched on the first try.
match_s0_4	Set 1: Match firsthand account	**a telling of an event by someone who was there**	2 points if matched on the first try.
match_s0_5	Set 1: Match personification	**giving human actions to something that is not human**	2 points if matched on the first try.
match_s1_1	Set 2: Match origin	**the point (0, 0), where the two axes meet**	2 points if matched on the first try.
match_s1_2	Set 2: Match capacity	**the amount a container can hold**	2 points if matched on the first try.
match_s1_3	Set 2: Match equivalent fractions	**fractions that name the same amount**	2 points if matched on the first try.
match_s1_4	Set 2: Match signal words	**words that point to a text structure**	2 points if matched on the first try.
match_s1_5	Set 2: Match turning point	**the moment a character begins to change**	2 points if matched on the first try.
match_s2_1	Set 3: Match quadrilateral	**a closed figure with four straight sides**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_2	Set 3: Match line of symmetry	**a fold line that makes two matching halves**	2 points if matched on the first try.
match_s2_3	Set 3: Match hyperbole	**a huge exaggeration not meant to be taken literally**	2 points if matched on the first try.
match_s2_4	Set 3: Match secondhand account	**a telling of an event by someone who was not there**	2 points if matched on the first try.
match_s2_5	Set 3: Match inference	**text clues plus what you already know**	2 points if matched on the first try.

UNIT 5 BOSS CHALLENGE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
z1_s0	Set 1: *The Situation In Pittsburgh, Ximena builds a model truss from equilateral triangles. Each triangle has a perimeter of 3 3/4 feet. The bottom beam of the model is made of 4 triangle sides in a row. That evening, Ximena writes about the day in her journal.* Three steps: how many INCHES long is the bottom beam? Inches:	**60** (also accepted: 60 inches, 60 inch, 60 in)	Step 1: an equilateral triangle has 3 equal sides: 3 3/4 ÷ 3 = 15/4 ÷ 3 = 5/4 = 1 1/4 feet. Step 2: 1 1/4 feet = 12 + 3 = 15 inches. Step 3: 4 sides in a row: 4 × 15 = 60 inches.
z2_s0	Set 1: Which journal sentence is a FIRSTHAND detail AND uses a simile? [I watched a slow boat slide under the bridge like a turtle. / The boat was a turtle under the bridge. / Pittsburgh has many bridges over its rivers. / Boats slide under bridges like turtles.]	**I watched a slow boat slide under the bridge like a turtle.**	It uses I (a firsthand detail from someone who was there) and compares the boat to a turtle using like (a simile). One choice is a metaphor with no I, one is a fact with no figurative language, and one is not firsthand.
z1_s1	Set 2: *The Situation In Duluth, Solveig measures 6 scraps of rope, in yards: 1/2, 3/4, 3/4, 1, 1 1/4, and 1 3/4. She plots them on a line plot, then ties them end to end. Later she writes a report about the tugboat.* Two steps: what is the total length of all 6 scraps, in FEET? Feet:	**18** (also accepted: 18 feet, 18 foot, 18 ft)	Step 1: total in fourths: 2/4 + 3/4 + 3/4 + 4/4 + 5/4 + 7/4 = 24/4 = 6 yards. Step 2: 1 yard = 3 feet, so 6 × 3 = 18 feet. (This ignores the rope used up in the knots.)
z2_s1	Set 2: Solveig's report says: The tugboat's engine kept overheating. To fix this, Uncle Torvald cleaned the cooling pipes, and the engine ran smoothly again. Which answer names the structure AND gives the best summary? [Problem and solution: an overheating engine ran smoothly after the pipes were cleaned. / Compare and contrast: an overheating engine ran smoothly after the pipes were cleaned. / Problem and solution: Uncle Torvald is the best deckhand in Duluth. / Problem and solution: the tugboat had an engine.]	**Problem and solution: an overheating engine ran smoothly after the pipes were cleaned.**	The paragraph names a problem (the engine overheats) and a solution (cleaning the pipes), so it is problem and solution, and the summary keeps both in new words. One choice names the wrong structure, one adds an opinion, and one leaves out the solution.

ID	Question	Answer	Why (explain to student)
z1_s2	Set 3: *The Situation In Sitka, Anya draws a map of the harbor on a grid. The dock is at (3, 2), the buoy is at (3, 9), and the island is at (8, 9). Each grid unit stands for 1/2 mile. Later she reads a story about a young sailor.* Three steps: Anya rides along the grid lines from the dock to the buoy to the island, then back the same way. How many miles does she ride in all? Miles:	**12** (also accepted: 12 miles, 12 mile, 12 mi)	Step 1: dock to buoy: 9 − 2 = 7 units. Buoy to island: 8 − 3 = 5 units. One way: 7 + 5 = 12 units. Step 2: 12 × 1/2 = 6 miles one way. Step 3: there and back: 6 × 2 = 12 miles.
z2_s2	Set 3: At the end of the story, the once-timid sailor says, “My heart was a drum, but I steered us home.” Which answer names the figurative language AND the change it shows? [A metaphor; he was nervous but became brave enough to act. / A simile; he was nervous but became brave enough to act. / A metaphor; he plays the drums on the boat. / Hyperbole; he was calm the whole time.]	**A metaphor; he was nervous but became brave enough to act.**	“My heart was a drum” says one thing IS another, so it is a metaphor; a pounding heart shows he was nervous. “But I steered us home” shows he acted bravely anyway. The others name the wrong kind or the wrong change.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Checkpoint_PrePost_G4-5W25.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Start at the origin, move 5 right and 3 up; is (0, 7) on the x-axis or the y-axis?	(5, 3). (0, 7) is on the y-axis, because its x-coordinate is 0.
2	Convert 2 1/3 yards to feet and 4.2 kilograms to grams	(a) 7 feet (2 × 3 = 6, and 1/3 of 3 = 1). (b) 4,200 grams (4.2 × 1,000).
3	Line plot: 2 X’s at 3/8 inch and 3 X’s at 1/2 inch; total length of the 5 shells	2 1/4 inches. 2 × 3/8 = 6/8 and 3 × 1/2 = 12/8; 6/8 + 12/8 = 18/8 = 2 2/8 = 2 1/4 inches.
4	Full name of a triangle with angles 90°, 45°, 45° and two equal sides; lines of symmetry in a square	A right isosceles triangle (accept isosceles right triangle). A square has 4 lines of symmetry.
5	“Because the trail was icy, the rangers closed it.” Signal word and text structure	Circle Because. It points to cause and effect (cause: the icy trail; effect: the rangers closed it).
6	Which sentence uses a simile?	The harbor was as quiet as a library. (It uses as. “The harbor was a library” is a metaphor.)

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	A 5-yard rope; Solveig cuts off 7 feet; how many INCHES are left?	96 inches. 5 yards = 15 feet; 15 − 7 = 8 feet; 8 × 12 = 96 inches. (Or 180 − 84 = 96 inches.)
2	Uzoma says 3 cups shared among 4 jars is 1 1/3 cups each	He divided the wrong way (4 ÷ 3). Share the total among the jars: 3 ÷ 4 = 3/4 cup in each jar. Check: 4 × 3/4 = 3 cups.

#	Prompt	Model answer / grading note
3	Be the editor	a) Aunt Winnie said, "Check every angle twice." b) On June 2, 2026, we visited Sitka, Alaska. c) The crabs in the pool hide under the rocks. (Accept hid; the subject is crabs, not pool.)
4	Writing: a firsthand account of a visit to one Unit 5 place	Grading note: written with I as if the student was there; names a Unit 5 place (for example, Yellowstone, the Duluth harbor, Haystack Rock, Pittsburgh, Mesa Verde, Nashville, or Sitka); a signal word followed by a comma; one simile or metaphor; one correctly punctuated line of dialogue; 4–5 complete sentences.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	A (2, 1) to B (2, 7) to C (6, 7); each grid unit is 1/4 mile	2 1/2 miles. A to B: 7 − 1 = 6 units. B to C: 6 − 2 = 4 units. 6 + 4 = 10 units; 10 × 1/4 = 10/4 = 2 2/4 = 2 1/2 miles.